

School of Statistics
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Academic Appointments

IRSA Faragher Distinguished Postdoctoral Fellow, School of Statistics, University of Minnesota
2025–2027

Education

Ph.D. in Statistics (2025), University of Minnesota,
Adviser: Adam J. Rothman
M.S. in Mathematical Sciences (2020), University of Minnesota,
B.S. in Statistics (2018), Jilin University, China

Research interests

Quantum Information Science (theory, advantage, and tomography); High-Dimensional Statistics; Statistical Learning and Generative AI; Astrophysics and Cosmology; Non-Convex Optimization.

Research

— Selected manuscripts submitted for publication

* Undergraduate research mentee under my supervision.

21. [Hongru Zhao](#). (2026) [Minimax Quantum State Tomography with Periodic Clifford Measurements](#). Preprint. [Lean verified](#). [GitHub](#).
20. [Hongru Zhao](#). (2026) [Local Anticoncentration for Gaussian Boson Sampling via Conditional Wishart Geometry](#). Preprint. [Lean verified](#). [GitHub](#).
18. [Hongru Zhao](#). (2026) [Uniform Hiding and Two Routes to Relative Accuracy in Gaussian Boson Sampling](#). Preprint. [Lean verified](#). [GitHub](#).
14. [Hongru Zhao](#). (2026) [On the Log Determinant of Sample Correlation Matrices under Gaussianity](#). Preprint.
12. [Hongru Zhao](#) and [Huiqian Feng](#)*. (2026) [Convex Reparameterization and Self-Concordant Algorithms for Multivariate Regression with Covariance Estimation](#). Preprint. [GitHub](#).
11. [Hongru Zhao](#), [Jinwen Fu](#), [Tuan Pham](#). (2025) [Convergence and Stability Analysis of Self-Consuming Generative Models with Heterogeneous Human Curation](#). Preprint.
10. [Hongru Zhao](#). (2025) [Allocation-Constrained Adaptive Design and Sequential Estimation: Theory and Application to Bradley–Terry–Davidson Models](#). Preprint. [GitHub](#).

— Peer reviewed articles

† Authors are listed alphabetically; author order does not indicate relative contribution.

9. Hongru Zhao, Aaron J. Molstad, and Adam J. Rothman. (2026) [Subspace Decompositions for Association Structure Learning in Multivariate Categorical Response Regression](#). Second-round minor revision at Electronic Journal of Statistics. [GitHub](#).
8. Hongru Zhao and Xiaotong Shen. (2026) [Distributed Algorithms for High-Dimensional Statistical Inference and Structure Learning with Heterogeneous Data](#). Statistica Sinica, forthcoming in 38(1) (January 2028).
7. Xiangyu Zhang, Erik Floden, [Hongru Zhao](#), Sara Algeri, Galin Jones, Vuk Mandic, and Jesse Miller. (2026) [Validating Angular Power Spectral Models for the Stochastic Gravitational-Wave Background Without Distributional Assumptions](#). Physical Review D, 113(2), 023051.
6. Sara Algeri, Xiangyu Zhang, Erik Floden, Hongru Zhao, Galin L. Jones, Vuk Mandic, and Jesse Miller. (2026) [Testing models for angular power spectra: A distribution-free approach](#). Physical Review D, 113(2), L021306.
5. Zhibo Zhang, Bixiong Li, Zhiwen Wang, Lianghui Li, Qingshun Nong, and [Hongru Zhao](#). (2026) [Microstructure-sensitive modeling of foam concrete based on pore-scale effect quantification and nonlinear assessment of its multifunctional performance](#). Construction and Building Materials, 537, 147017.
4. Xiaou Li and [Hongru Zhao](#)†. (2025) [Globally-Optimal Greedy Active Sequential Estimation](#). IEEE Transactions on Information Theory, 71(5), 3871–3924. [GitHub](#).
3. [Hongru Zhao](#), and Jinchao Xu. (2024) [Convergence Analysis and Trajectory Comparison of Gradient Descent for Overparameterized Deep Linear Networks](#). Transactions on Machine Learning Research, ISSN: 2835-8856. [GitHub](#).
2. Kangsheng Liu, Zhuangyi Liu, and Hongru Zhao†. (2024) [Exponential stability of the linear KdV-BBM equation](#). Discrete and Continuous Dynamical Systems-B, 29(3), 1206-1216.
1. Yongcheng Qi, and [Hongru Zhao](#)†. (2021) [Limiting empirical spectral distribution for products of rectangular matrices](#). Journal of Mathematical Analysis and Applications, 502(2),125237.

— Additional manuscripts and preprints

19. [Hongru Zhao](#). (2026) [Shifted Anticoncentration for Real Gram Hafnians and Symmetric Gaussian Hafnians](#). Preprint. [Lean verified](#). [GitHub](#).
17. [Hongru Zhao](#). (2026) [Weak Typicality of von Neumann Entanglement Entropy in Gaussian Boson Sampling](#). Preprint. [Lean verified](#).
16. [Hongru Zhao](#). (2026) [Exact Moments of Gaussian Gram Hafnians Reveal an \$n^2/\log n\$ Threshold for Weak Anticoncentration](#). Preprint. [Lean verified](#).
15. [Hongru Zhao](#). (2026) [Sharp Berry–Esseen Bounds for the Log Determinant of a Gaussian Sample Correlation Matrix](#). Preprint. [Lean verified](#). [GitHub](#).
13. [Hongru Zhao](#) and Galin Jones. (2026) [A Counterexample to Karlin’s Conjectured Converse for Admissibility in Exponential Families](#). Preprint.

Mentoring & Supervision

- Undergraduate Research
Huqian Feng* — Undergraduate Research Mentee (high-dimensional statistics),
School of Statistics, UMN Fall 2025 – Spring 2026

Invited presentations

Invited Talk and Scholarly Exchange, School of Physics and Astronomy, Sun Yat-sen University, China: Statistical Inference for the Statistical Isotropy Stochastic Gravitational Wave Background (2026).

Invited Talk, Center for Applied Mathematics, Tianjin University, China: From Fisher Information to Adaptive Experimental Design: Statistical Optimality in Quantum State Tomography (2026).

Contributed Paper Presentation, Joint Statistical Meetings (JSM), Nashville, TN: Subspace Decompositions for Association Structure Learning in Multivariate Categorical Response Regression (2025).

Poster Presentation, IRSA Conference, School of Statistics, University of Minnesota: Structure Learning in Multivariate Categorical Response Regression (2024).

Colloquium Seminar, Computer, Electrical, and Mathematical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia: Exploring the Optimization Landscape and Training Dynamics of Deep Linear Networks (2023).

Professional service

- Departmental Service
Seminar Coordinator, UMN School of Statistics Seminar Series, Fall 2025
- External Outreach and Sponsorship
Sponsor representative, WiADS Conference — Represented the IRSA; staffed the sponsor desk, and promoted IRSA consulting and statistical workshops.
- Reviewer Service
Journal of Machine Learning Research (2026); Journal of the American Statistical Association (2026); IEEE Transactions on Information Theory (2025); Electronic Journal of Statistics (2025); Annals of Applied Probability (2022); Transactions on Machine Learning Research (2024–2025).

Courses taught

- UMN-TC: Graduate Instructor
Spring 2026: STAT 4052 Statistical Machine Learning II
Fall 2025: STAT 4102 Theory of Statistics II
Spring 2025: STAT 3301 Regression and Statistical Computing
Fall 2024: STAT 4101 Theory of Statistics I
Spring 2024: STAT 3301 Regression and Statistical Computing
Fall 2023: STAT 3011 Introduction to Statistical Analysis
- UMN-TC: Graduate Teaching Assistant
Fall 2022: STAT 5101 Theory of Statistics I
Spring 2022: STAT 5102 Theory of Statistics II
Fall 2021: STAT 3021 Introduction to Probability and Statistics

Programming Skills

Lean, Mathematica, MATLAB, Python, R.